

UEI403: ELECTRICAL AND ELECTRONIC MEASUREMENTS				
	L	T	P	Cr
	3	1	2	4.5
Course Objective: To understand concepts of various electrical and electronic measuring instruments. To familiarize with different electromechanical and electronic instruments. To introduce instruments for power and energy measurements. To explain instrument transformers and magnetic measurements. To be able to measure different physical parameters with the help of AC bridges.				
Syllabus Electrical Standards: Types of standards, standards of E.M.F. and resistance, Frequency dependence of resistance, Inductance and Capacitance, Time and frequency standards. Electromechanical Indicating Instruments: PMMC galvanometer, Ohmmeter, Electrodynamometer, Moving iron meter, Rectifier and thermo-instruments, Comparison of various types of indicating instruments. Power and Energy Measurement: Electrodynamometer type of wattmeter and power factor meter, Power in poly phase system: two wattmeter method, Single-phase induction and Electronic energy meters. Instrument Transformers: Current & Voltage transformers, Constructional features, Ratio & Phase angle errors. Magnetic Measurements: Determination of B-H curve and hysteresis loop, Measurement of iron losses with Llyod Fisher square. Bridge Measurements: AC bridges: Applications and conditions for balance, Maxwell's bridge, Hay's bridge, Schering Bridge, Wien's bridge, De Sauty's bridge, Anderson's Bridge, Insulation testing, Ground resistance measurement, Varley and Murray loop test. Electronic Instruments: Electronic multimeter, Digital voltmeters, General characteristics ramp type voltmeter, Quantization error, Digital frequency meter/Timer, Q meter and its applications, Distortion meter, Wave meter and Spectrum Analyzer, Block diagram and Applications of oscilloscopes, Deflection sensitivity, Frequency measurement using Lissajous figures, Storage type digital oscilloscopes.				
Laboratory Work: Experiments around sensitivity of wheat stone bridge, Comparison of various types of indicating instruments, Single-phase induction type energy meter, AC bridges, Measurement of iron losses with Llyod Fisher square, Storage type digital oscilloscopes.				
Course Learning Objectives (CLO) The students will be able to: 1. Exhibit the knowledge of working of different electromechanical indicating instruments. 2. Measure power and energy with the help of wattmeter and energy meter. 3. Exhibit the knowledge of working and applications of instrument transformers and magnetic measurements. 4. Elucidate the working and application of several AC bridges for inductance and capacitance.				

5. Exhibit the knowledge of working principle and applications of various electronic instruments.

Text Books

1. Electrical Measurements and Measuring Instruments, Golding, E.W., and Widdis, F.C., Pitman (2003).
2. Modern Electronic Instrumentation and Measurement Techniques, Helfrick, A.D., and Cooper, W.D., Prentice Hall of India (2007).

Reference Books

1. Electronic Instrumentation, Kalsi, H.S., Tata McGrawHill (2007).
2. Instrumentation Measurement and Analysis, Nakra, B.C., Chaudhry, K.K., Tata McGrawHill (2003).

Evaluation Scheme	
Evaluation Elements	Weightage %
Mid Semester Test (MST)	25-30
End Semester Examination (ESE)	40-45
Sessional (may include Assignment, Sessional (Includes Regular Lab assessment) and Quizzes Project (Including report, presentation etc.)	30